

Parsa Esmkhani

709-986-6515 | pesmkhani@mun.ca | parsaesmkhani.github.io | linkedin.com/in/parsaesmkhani

EDUCATION

B.Sc. (Honours), Major in Computer Science, Minor in Mathematics Sep 2021 – Dec 2026 (Exp.)
Memorial University of Newfoundland *St. John's, NL*

- GPA: 3.97/4.00 | Concentration: Theory of Computation
- Interests: Social Choice Theory, Game Theory, Logic, Complexity Theory, Cognition, Fairness Theory, Graph Theory

RESEARCH EXPERIENCE

Summer Undergraduate Research Award (SURA) May 2026 – Present
Faculty of Medicine *Memorial University*

- Developing a Python transcription workflow and SQLite-backed database to transform 78 qualitative interviews into a searchable research corpus, reducing manual data preparation and enabling downstream machine-learning-based information extraction under Dr. Michael Wahl.

Research Assistantships Sep 2025 – Present
Faculties of Geography, Economics, and Business *Memorial University*

- **Geography** (Jun 2026–Present; Dr. Nicholas Lynch): Collecting and synthesizing literature and web-scraped data to support research on demolition-material reuse, expanding the evidence base for circular-economy analyses.
- **Mathematical Economics** (May 2026–Present; Dr. Jean-Gabriel Lauzier): Integrating literature metadata into graph-based knowledge systems (Obsidian) to visualize citation relationships and streamline literature navigation.
- **Business** (Sep 2025–May 2026; Dr. Chansoo Park, Dr. Phillip Piercey): Developed an AI-assisted screening pipeline that reduced 38,000+ Factiva articles to a validated corpus of 4,800+ records, dramatically reducing manual review while producing a research-ready dataset for empirical and sentiment analyses of divestment media coverage.

Honours Research Thesis | Game Theory, Complexity Theory Fall 2025 – Winter 2026

- Proved a derandomization result showing that, under suitable assumptions, if a pseudo-random generator exists, Reality can keep Skeptic's capital bounded in a variant of the Skeptic–Reality game in game theoretic probability.
- Supervised by Dr. Antonina Kolokolova.

PRESENTATIONS

- **Familiarity in Humans vs. Artificial Neural Networks** — Centre for Vision Research Summer School, York University (Jun 12, 2026)
- **Derandomization in a Skeptic–Reality Game** — SEA Conference, Memorial University (Apr 9, 2026)
- **Fairness in Resource Allocation; Cake Cutting** — Theory Group Seminar, Memorial University (Mar 11, 2025)

TEACHING EXPERIENCE

Peer Assisted Learning [PAL] Leader for COMP 1002 (Logic) Winter 2023 – Present
Academic Success Center *Memorial University*

- Conducted two weekly PAL sessions for 11 semesters, and serving as a mentor to other PAL leaders for 3.
- Accumulated the highest number of student contact hours among all PAL subjects, totaling 1,700+ hours.

Teaching Assistant — COMP 1002: Introduction to Logic Winter 2026
Department of Computer Science *Memorial University*

- Provided instructional support for ~60 students during weekly labs, as well as grading and exam invigilation.

Computer Science Help Centre Assistant Fall 2025/Winter 2026
Department of Computer Science *Memorial University*

- Provided support for Computer Science students in year 1 and 2 for COMP 1001, 1002, 1003, 2001, 2002, and 2003

Substitute Instructor — COMP 1002: Introduction to Logic Fall 2025/Winter 2026
Department of Computer Science *Memorial University*

- Delivered four lectures in place of the course instructor on induction, structural induction, FOL logic, and relations.

Residence Computer Science/Mathematics Assistant Fall 2025
Hatcher Commons – Student Residences *Memorial University*

- Provided on-site assistance for residence students in first-year Computer Science and Mathematics courses.

OTHER WORK EXPERIENCE

Persian/Dari Interpreter

Oct 2021 – Jan 2024 | Feb 2024 – Present

Association for New Canadians [ANC] | Kelly Services Canada

St. John's, NL | Remote

- Trained by Language Line Solutions; 8,000+ hours of medical, banking, and educational interpretation experience.

World University Service of Canada [WUSC] Ambassador

Fall 2021 – Spring 2022

Internationalization Office, Memorial University of Newfoundland

St. John's, NL

- Assisted the local WUSC scholar with settlement in Canada and organized events for the WUSC MUN chapter.

AWARDS & RESEARCH FUNDING

- **1st Place, IBM watsonx Hackathon (2026)** — Winning team among 16 teams for *B'y Your Side*
- **Accepted to Carnegie Mellon University (CMU) Summer School in Logic and Formal Epistemology (2026)** — One of 25 students selected globally; unable to attend due to U.S. visa restrictions affecting Iranians
- **Accepted to York University Centre for Vision Research (CVR) Vision Science Summer School (2026)** — One of 20 students selected globally; travel and accommodation provided by York University
- **Summer Undergraduate Research Award (SURA) (2026)** — Received \$3,000 in funding to conduct research under the supervision of Dr. Michael Wahl
- **ISM Discovery School: Complexity and Mathematical Logic Travel Funding (2026)** — Received \$1,500 in travel funding to attend the ISM Discovery School at *Concordia University* in Montreal
- **Computational Complexity Conference Travel Funding (2025)** — Received \$2,100 in travel funds (\$300 from CCC, \$1,800 from supervisor) to attend CCC at the *Fields Institute, University of Toronto*
- **Dean's List Award (2023–2024) and (2024–2025)** — Top 10% of Faculty of Science students based on GPA

PROJECTS

IntuitiveCS.ca | *Python, ReactFlow, CSS, HTML, JavaScript*

Dec 2025 – Present

- Co-developed a website featuring interactive visualizations of mathematical computer science concepts to support instruction and student learning in logic and intro theoretical computer science courses.

B'y Your Side | *IBM watsonx, Healthcare AI, LLMs, Wearables*

May 29–31, 2026

- Co-developed a context-aware healthcare AI assistant that integrates patient records and wearable-vital data to personalize health alerts using medications, diagnoses, risk factors, and historical health trends ([Video Demo](#)).

Game of Four | *Java, Java Swing, JUnit*

Fall 2023

- Co-developed a desktop implementation of the Game of Four board game for COMP 2005 ([Video Demo](#)).

COMP 1002 Interactive Logic Exercises | *Carnap, Markdown*

Spring 2022

- Assisted Dr. McIntyre in developing proof-checking tools and interactive exercises ([Repository](#)).

ACADEMIC LEADERSHIP & SERVICE

- **Student Rep, Department Head Review Committee** | *Department of Computer Science* May 2026 - Present
- **Undergraduate Committee Student Rep** | *Department of Computer Science* Sep 2025 – Present
- **Co-Organiser of Theory Group** | *Department of Computer Science* Sep 2023 – Present
- **Science Student Rep, Annual Student Recruitment Panel** | *Faculty of Science* Feb 2026
- **Faculty of Science Student Rep for Computer Science** | *Faculty of Science* Jan 2024 – May 2025
- **Computer Science Society Executive (Social Outreach)** | *Memorial University* Jan 2022 – Dec 2025

VOLUNTEERING

- **Everything Science Volunteer** | *Memorial University, St. John's, NL* Jan 2024 – Present
- **SEA Conference Volunteer** | *Memorial University, St. John's, NL* Apr 2024, 2025, 2026
- **Canada Games 2025 Volunteer (Venue Technology Crew)** | *Canada Games, St. John's, NL* Aug 2025
- **Sponsorship Lead, Hack Frost NL** | *St. John's, NL* Sep 2022 – Feb 2024